

Junior Developer

Task Brief

Social Media Image Resizer Tool

Build a free, in-browser image resizer for social media platforms — a natural companion to the AiTechies BG Remover. No backend. No upload. 100% client-side.

■ Project	aitechies-image-resizer	■ Timeline	5 – 7 Days
■ Deploy	imgresizer.aitechies.in (Netlify)	■■■■ Assigned To	Junior Developer
■ Stack	Next.js 14 · Tailwind CSS · Canvas API	■ Repo	Create new — aitechies-image-resizer

■ Why This Project

- BG Remover complement — users remove BG then need to resize. Same workflow, same audience.
- High SEO value — 'free image resizer' is a high-search-volume keyword that drives organic traffic.
- Company portfolio — adds another production tool under the AI Techies free tools suite.
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- Junior-appropriate scope — Canvas API + Next.js. No AI/ML complexity, but real production pattern.
- Client utility — useful for Carspetta, WorkAxis, and all client social media work.

■ Platform Presets to Support

All presets must be hardcoded in `presets.ts` and displayed as selectable cards in the UI.

Platform	Preset Name	Width	Height
Instagram	Post (Square)	1080 px	1080 px
Instagram	Story / Reel Cover	1080 px	1920 px
Instagram	Landscape Post	1080 px	566 px
LinkedIn	Post	1200 px	627 px
LinkedIn	Profile Banner	1584 px	396 px

Platform	Preset Name	Width	Height
Twitter / X	Post	1200 px	675 px
Twitter / X	Header	1500 px	500 px
Facebook	Cover Photo	820 px	312 px
Facebook	Post	1200 px	630 px
YouTube	Thumbnail	1280 px	720 px
WhatsApp	Display Picture	500 px	500 px
Custom	User-defined	— px	— px

■ ■ Core Features

1. Upload & Preview	Drag-and-drop or click-to-browse. Accept PNG, JPG, WEBP. Show instant canvas preview of the image before resize.
2. Platform Picker	Grouped by platform (Instagram, LinkedIn, Twitter, etc.). Each preset shown as a card with dimensions label. Custom option allows manual W × H input.
3. Resize Modes	Fit — Letterbox, add padding. Show background colour picker. Fill — Centre-crop, no empty space. Stretch — Force exact dimensions.
4. Live Preview	Canvas updates in real-time as user switches presets or modes. Show output dimensions below canvas.
5. Download	Export as PNG / JPG / WebP. JPG/WebP shows a quality slider (60–100%). Filename auto-set: original-name_platform_WxH.ext
6. Batch (Stretch Goal)	Apply same settings to multiple uploaded images at once. Download as individual files or ZIP.

■ Folder Structure

Follow this exact structure. Do not deviate.

```
/app
├── index.html
├── src
│   ├── components
│   │   ├── MainUI
│   │   │   ├── MainUIEntryPoint
│   │   │   └── MainUIContent
│   │   ├── UploadZone
│   │   ├── PlatformSelector
│   │   ├── LiveCanvas
│   │   └── ResizeModes
│   └── utils
│       ├── fileUtils
│       └── imageUtils
└── assets
```

← Format + quality + download

← Core Canvas resize logic

← All platform dimensions

← Blob to file download helper

← Platform metadata + icons

■ Key Code Reference

presets.ts — Platform dimensions array

```
export const
  "Instagram"
  "Post (Square)"
  "Story / Reel"
  "Landscape"

  "LinkedIn"
  "Post"
  "Profile Banner"

// ... add Twitter, Facebook, YouTube, WhatsApp, Custom
```

resizeImage.ts — Core Canvas resize function

```
export function

  "fit"  "fill"  "stretch"
  "#ffffff"

return new
const      new

const      "canvas"

const      "2d"
if      "fit"

const
const
```

```
const
    else if "fill"
const
const
const

else

"image/png"
```

User Flow

Step 1	Upload Image Drag-and-drop or click. PNG / JPG / WEBP accepted.
Step 2	Pick Platform Instagram / LinkedIn / Twitter / Custom — card-based selector.
Step 3	Pick Preset Post / Story / Banner — show exact dimensions.
Step 4	Choose Mode Fit (with BG colour) · Fill (centre-crop) · Stretch.
Step 5	Live Preview Canvas updates instantly. Show output W × H below.
Step 6	Download Select PNG / JPG / WebP · Set quality · Click Download.

Branding Rules

Match the BG Remover site exactly. This must look like part of the same product suite.

Header	AI Techies · logo top-left · same nav as BG Remover
Footer	Built by AiTechies · Ansar Ibrahim — exact same footer links
Font	Inter (Google Fonts) — same as BG Remover
Colours	Primary #6C47FF · Accent #00C2A8 · BG #0D0D0D (dark mode default)
Favicon	Use aitechies.in favicon — do not create a new one

Domain	Deploy to imgresizer.aitechies.in via Netlify
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■ Submission Checklist

Junior must verify all items before marking task as Done.

- All 11 platform presets working with correct pixel dimensions
- Fit mode — no image cut, padding fills correctly with chosen colour
- Fill mode — no empty space, image centred-cropped
- Stretch mode — forces exact $W \times H$
- Custom dimension input working (manual $W \times H$)
- Download works for PNG, JPG, and WebP formats
- JPG / WebP quality slider functional (60–100%)
- Filename auto-generated: originalname_platform_WxH.ext
- Mobile responsive — tested on 375px viewport
- Same branding as BG Remover — header, footer, colours, font
- Deployed live on Netlify (share preview URL with Ansar)
- Code pushed to GitHub repo — clean commits, no console errors

■ Suggested Timeline

Day 1	Project setup — Next.js 14, Tailwind, folder structure, presets.ts
Day 2	ImageUploader + PlatformPicker components
Day 3	resizeImage.ts core logic — Fit / Fill / Stretch
Day 4	CanvasPreview + live update wiring
Day 5	DownloadPanel — PNG/JPG/WebP + quality slider
Day 6	Branding polish, mobile responsive, QA
Day 7	Deploy to Netlify, final testing, submit